

Functions

ANSWER KEY



Evaluating functions

- 1 If $f(x) = 3x - 4$, find $f(5)$.

$$f(5) = 3(5) - 4 = 11.$$

- 2 If $f(x) = x^2 - 2x + 1$, find $f(-2)$.

$$f(-2) = 4 + 4 + 1 = 9.$$

- 3 If $g(x) = \sqrt{x + 5}$, find $g(4)$.

$$g(4) = \sqrt{9} = 3.$$

- 4 If $h(x) = \frac{2x}{x-1}$, find $h(3)$.

$$h(3) = \frac{6}{2} = 3.$$

Domain and range

- 5 State the domain of $f(x) = \sqrt{x - 3}$.

$$x - 3 \geq 0, \text{ so the domain is } x \geq 3.$$

- 6 State the domain of $f(x) = \frac{1}{x-5}$.

All real numbers except $x = 5$ (where the denominator is zero).

- 7 State the domain of $f(x) = \sqrt{7 - x}$.

$$7 - x \geq 0, \text{ so the domain is } x \leq 7.$$

- 8 State the range of $f(x) = x^2 + 2$.

Since $x^2 \geq 0$ for all real x , the range is $f(x) \geq 2$.

Transformations: shifts

- 9 The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.

$$y = (x - 3)^2 - 2.$$

- 10 The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.

$$y = |x + 4| + 1.$$

- 11 Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.

A shift 2 units left and 5 units down.

Transformations: reflections and stretches

- 12 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 3. Write the new equation.

$$y = -3x^2.$$

- 13 Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.

A vertical stretch by a factor of 2.

- 14 Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.

A reflection over the x -axis, then a shift 1 unit right and 4 units up.

Composite functions

- 15 Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.

$$g(2) = 4 - 3 = 1. f(1) = 2(1) + 1 = 3.$$

- 16 Using the same f and g , find $(g \circ f)(2)$.

$$f(2) = 5. g(5) = 25 - 3 = 22.$$

- 17 Using the same f and g , find a general formula for $(f \circ g)(x)$.

$$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5.$$

- 18 Using the same f and g , find a general formula for $(g \circ f)(x)$.

$$g(f(x)) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3 = 4x^2 + 4x - 2.$$

Inverse functions

- 19 Find the inverse of $f(x) = 3x - 2$.

$$y = 3x - 2 \Rightarrow x = \frac{y + 2}{3}. \text{ So } f^{-1}(x) = \frac{x + 2}{3}.$$

- 20 Find the inverse of $f(x) = \frac{x-4}{5}$.
 $y = \frac{x-4}{5} \Rightarrow 5y = x-4 \Rightarrow x = 5y+4$. So $f^{-1}(x) = 5x+4$.
- 21 Show that $f(x) = 2x+6$ and $g(x) = \frac{x-6}{2}$ are inverses by computing $f(g(x))$.
 $f(g(x)) = 2\left(\frac{x-6}{2}\right) + 6 = (x-6) + 6 = x$, confirming they are inverses.
- 22 Find the inverse of $f(x) = x^3 + 1$.
 $y = x^3 + 1 \Rightarrow x = y^3 + 1 \Rightarrow y = \sqrt[3]{x-1}$. So $f^{-1}(x) = \sqrt[3]{x-1}$.
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Piecewise functions

- 23 A function is defined by $f(x) = x+2$ for $x < 1$, and $f(x) = 3x-1$ for $x \geq 1$. Find $f(-2)$.
Since $-2 < 1$, use the first rule: $f(-2) = -2+2 = 0$.
- 24 Using the same piecewise function, find $f(1)$.
Since $1 \geq 1$, use the second rule: $f(1) = 3(1)-1 = 2$.
- 25 Using the same piecewise function, find $f(4)$.
Since $4 \geq 1$, use the second rule: $f(4) = 3(4)-1 = 11$.
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Even and odd functions

- 26 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.
 $f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, so f is even.
- 27 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.
 $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$, so f is odd.
- 28 Determine whether $f(x) = x^2 + x$ is even, odd, or neither.
 $f(-x) = x^2 - x$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.
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Increasing, decreasing, and average rate of change

- 29 Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.
 $\frac{f(4) - f(1)}{4 - 1} = \frac{16 - 1}{3} = \frac{15}{3} = 5$.

- 30 Find the average rate of change of $f(x) = 2x^2 - 3x$ from $x = 0$ to $x = 3$.

$$f(3) = 18 - 9 = 9, f(0) = 0. \frac{9 - 0}{3} = 3.$$

- 31 State the interval on which $f(x) = (x - 2)^2$ is decreasing.

The vertex is at $x = 2$ and the parabola opens upward, so f is decreasing for $x < 2$.

Operations on functions

- 32 Let $f(x) = x + 3$ and $g(x) = x^2 - 1$. Find $(f + g)(x)$.

$$(f + g)(x) = (x + 3) + (x^2 - 1) = x^2 + x + 2.$$

- 33 Using the same f and g , find $(f \cdot g)(2)$.

$$f(2) = 5, g(2) = 3. (f \cdot g)(2) = 5 \times 3 = 15.$$

- 34 Using the same f and g , find $(f/g)(3)$.

$$f(3) = 6, g(3) = 8. (f/g)(3) = \frac{6}{8} = \frac{3}{4}.$$

Reading a function graph

- 35 The graph of $y = f(x) = x^2 - 4$ is shown.

- a State the zeros of f . $x = -2$ and $x = 2$ (where the curve crosses the x -axis).
 - b State the domain and range of f . Domain: all real numbers. Range: $y \geq -4$.
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Word problems: functions in context

- 36 This question has two parts. A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 150$.

- a Evaluate $P(10)$ to find the profit from selling 10 units.

$$P(10) = -2(100) + 800 - 150 = -200 + 800 - 150 = 450 \text{ dollars.}$$

- b Find the number of units x that maximizes profit (using the vertex of the parabola), and state the maximum profit. The vertex is at $x = -\frac{80}{2(-2)} = 20$ units. $P(20) = -2(400) + 1600 - 150 = 650$ dollars — the maximum profit.

- 37 This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city t hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \leq t \leq 12$.
- a Find $C(4)$, the Celsius temperature 4 hours after 6 a.m.
 $C(4) = -0.5(16) + 24 + 10 = -8 + 24 + 10 = 26^\circ\text{C}$.
- b Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.
 $(F \circ C)(4) = \frac{9}{5}(26) + 32 = 46.8 + 32 = 78.8^\circ\text{F}$.
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MCQ — Evaluating functions

- 38 If $f(x) = 4x - 7$, find $f(6)$.

a. 17 ✓

b. 24

c. 31

d. 10

$$4(6) - 7 = 24 - 7 = 17.$$

- 39 If $f(x) = x^2 - 3x + 2$, find $f(-3)$.

a. 2

b. -16

c. 20 ✓

d. 16

$$(-3)^2 - 3(-3) + 2 = 9 + 9 + 2 = 20.$$

40 If $g(x) = \sqrt{x + 9}$, find $g(7)$.

a. 8

b. 16

c. 4 ✓

d. $\sqrt{16}$

$$g(7) = \sqrt{16} = 4.$$

41 If $h(x) = \frac{3x}{x - 2}$, find $h(5)$.

a. 15

b. 5 ✓

c. 3

d. 1.5

$$h(5) = \frac{15}{3} = 5.$$

42 If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

a. 18

b. 8

c. 16 ✓

d. 22

$$f(3) = 19, f(1) = 3; 19 - 3 = 16.$$

MCQ — Domain and range

43 State the domain of $f(x) = \sqrt{x - 5}$.

a. $x > 5$

b. $x \geq 5$ ✓

c. $x \leq 5$

d. $x \neq 5$

The expression under the root must be ≥ 0 : $x - 5 \geq 0 \Rightarrow x \geq 5$.

44 State the domain of $f(x) = \frac{1}{x - 7}$.

a. All reals except $x = 7$ ✓

b. $x > 7$

c. $x \geq 7$

d. All real numbers

The denominator cannot be 0, so $x \neq 7$.

45 State the domain of $f(x) = \sqrt{9 - x}$.

a. $x \leq 9$ ✓

b. $x < 9$

c. $x \neq 9$

d. $x \geq 9$

$$9 - x \geq 0 \Rightarrow x \leq 9.$$

46 State the range of $f(x) = x^2 + 5$.

a. $y > 5$

b. All real numbers

c. $y \leq 5$

d. $y \geq 5$ ✓

Since $x^2 \geq 0$, the minimum value of f is 5: range $y \geq 5$.

47 The graph shows $f(x)$ defined only where plotted. State its domain.

a. $[-3, 3]$ ✓

b. All real numbers

c. $[-4, 4]$

d. $[0, 3]$

The curve is only plotted for x between -3 and 3 .

MCQ — Transformations: shifts

48 The graph of $y = x^2$ is shifted 5 units right and 3 units down. Write the new equation.

a. $y = (x - 5)^2 + 3$

b. $y = (x - 5)^2 - 3$ ✓

c. $y = (x + 5)^2 - 3$

d. $y = (x - 3)^2 - 5$

Right 5: replace x with $x - 5$; down 3: subtract 3.

49 The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

a. $y = |x + 4| + 2$

b. $y = |x + 2| + 4$ ✓

c. $y = |x + 2| - 4$

d. $y = |x - 2| + 4$

Left 2: replace x with $x + 2$; up 4: add 4.

50 Describe the transformation that takes $y = x^2$ to $y = (x - 3)^2 + 2$.

- a. Shift right 2, up 3
- b. Shift right 3, down 2
- c. Shift left 3, up 2
- d. Shift right 3, up 2 ✓

Replacing x with $x - 3$ shifts right 3; adding 2 shifts up 2.

51 The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

- a. $y = \sqrt{x + 6}$
- b. $y = \sqrt{x} - 6$
- c. $y = \sqrt{x} + 6$
- d. $y = \sqrt{x - 6}$ ✓

Right 6: replace x with $x - 6$.

MCQ — Transformations: reflections and stretches

52 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 2. Write the new equation.

- a. $y = -\frac{1}{2}x^2$
- b. $y = (-2x)^2$
- c. $y = -2x^2$ ✓
- d. $y = 2x^2$

Reflect over x -axis: negate; stretch by 2: multiply by 2.

53 Describe the transformation that takes $y = \sqrt{x}$ to $y = 3\sqrt{x}$.

- a. Reflection over the x -axis
- b. Vertical shift up 3
- c. **Vertical stretch by a factor of 3 ✓**
- d. Horizontal stretch by a factor of 3

Multiplying the output by 3 stretches the graph vertically by a factor of 3.

54 Describe the transformation that takes $y = x^2$ to $y = -(x + 2)^2 + 1$.

- a. Reflect over x -axis, shift right 2, down 1
- b. Reflect over y -axis, shift right 2, up 1
- c. **Reflect over x -axis, shift left 2, up 1 ✓**
- d. Shift left 2, up 1 (no reflection)

The negative sign reflects over the x -axis; $+2$ shifts left 2; $+1$ shifts up 1.

55 The graph of $y = \sqrt{x}$ is reflected over the y -axis. Write the new equation.

- a. $y = -\sqrt{-x}$
- b. $y = -\sqrt{x}$
- c. **$y = \sqrt{-x}$ ✓**
- d. $y = \sqrt{x}$

Reflecting over the y -axis replaces x with $-x$: $y = \sqrt{-x}$.

MCQ — Composite functions

56 Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. Find $(f \circ g)(3)$.

a. 26 ✓

b. 23

c. 11

d. 29

$$g(3) = 8; f(8) = 3(8) + 2 = 26.$$

57 Using the same f and g , find $(g \circ f)(3)$.

a. 100

b. 120 ✓

c. 121

d. 26

$$f(3) = 11; g(11) = 11^2 - 1 = 120.$$

58 Using the same f and g , find a general formula for $(f \circ g)(x)$.

a. $3x^2 - 1$ ✓

b. $9x^2 - 1$

c. $3x^2 - 3$

d. $3x^2 + 2$

$$f(g(x)) = 3(x^2 - 1) + 2 = 3x^2 - 1.$$

59 Using the same f and g , find a general formula for $(g \circ f)(x)$.

a. $9x^2 + 12x + 3$ ✓

b. $3x^2 + 2x - 1$

c. $9x^2 + 4$

d. $9x^2 + 12x - 1$

$$g(f(x)) = (3x + 2)^2 - 1 = 9x^2 + 12x + 4 - 1 = 9x^2 + 12x + 3.$$

60 Let $f(x) = x + 4$ and $g(x) = 2x$. Find $(f \circ g)(5)$.

a. 10

b. 9

c. 14 ✓

d. 18

$$g(5) = 10; f(10) = 14.$$

MCQ — Inverse functions

61 Find the inverse of $f(x) = 4x - 3$.

a. $f^{-1}(x) = \frac{x-3}{4}$

b. $f^{-1}(x) = \frac{x+4}{3}$

c. $f^{-1}(x) = \frac{x+3}{4}$ ✓

d. $f^{-1}(x) = 4x + 3$

$$y = 4x - 3 \Rightarrow x = \frac{y+3}{4}.$$

62 Find the inverse of $f(x) = \frac{x-7}{2}$.

a. $f^{-1}(x) = 2x - 7$

b. $f^{-1}(x) = 2(x - 7)$

c. $f^{-1}(x) = 2x + 7$ ✓

d. $f^{-1}(x) = \frac{x+7}{2}$

$$y = \frac{x-7}{2} \Rightarrow x = 2y + 7.$$

63 Show that $f(x) = 3x + 9$ and $g(x) = \frac{x-9}{3}$ are inverses by computing $f(g(x))$.

a. x ✓

b. $x + 9$

c. $3x$

d. $x - 9$

$$f(g(x)) = 3\left(\frac{x-9}{3}\right) + 9 = x - 9 + 9 = x.$$

64 Find the inverse of $f(x) = x^3 - 2$.

a. $f^{-1}(x) = \sqrt[3]{x} + 2$

b. $f^{-1}(x) = (x + 2)^3$

c. $f^{-1}(x) = \sqrt[3]{x - 2}$

d. $f^{-1}(x) = \sqrt[3]{x + 2}$ ✓

$$y = x^3 - 2 \Rightarrow x = \sqrt[3]{y + 2}.$$

65 Find the inverse of $f(x) = 5x + 10$.

a. $f^{-1}(x) = \frac{x-10}{5}$ ✓

b. $f^{-1}(x) = \frac{x+10}{5}$

c. $f^{-1}(x) = \frac{x}{5} - 10$

d. $f^{-1}(x) = 5x - 10$

$$y = 5x + 10 \Rightarrow x = \frac{y-10}{5}.$$

MCQ — Piecewise functions

66 A function is defined by $f(x) = x + 3$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $f(-3)$.

a. -7

b. 0 ✓

c. 6

d. 1

$$x = -3 < 2: f(-3) = -3 + 3 = 0.$$

67 Using the same piecewise function, find $f(2)$.

a. 3 ✓

b. 4

c. 0

d. 5

$$x = 2 \geq 2: f(2) = 2(2) - 1 = 3.$$

68 Using the same piecewise function, find $f(5)$.

a. 5

b. 9 ✓

c. 7

d. 8

$$x = 5 \geq 2: f(5) = 2(5) - 1 = 9.$$

69 A function is defined by $f(x) = 2x$ for $x \leq 0$, and $f(x) = x^2 + 1$ for $x > 0$. Find $f(0) + f(3)$.

a. 1

b. 11

c. 0

d. 10 ✓

$$f(0) = 0; f(3) = 9 + 1 = 10; \text{sum} = 10.$$

MCQ — Even and odd functions

70 Determine whether $f(x) = x^2 - 7$ is even, odd, or neither.

- a. Neither
- b. Even ✓**
- c. Odd
- d. Both even and odd

$$f(-x) = x^2 - 7 = f(x): \text{even.}$$

71 Determine whether $f(x) = x^3 - 4x$ is even, odd, or neither.

- a. Neither
- b. Both even and odd
- c. Even
- d. Odd ✓**

$$f(-x) = -x^3 + 4x = -f(x): \text{odd.}$$

72 Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

- a. Both even and odd
- b. Neither ✓**
- c. Odd
- d. Even

$$f(-x) = x^2 - 3x, \text{ which is neither } f(x) \text{ nor } -f(x): \text{neither.}$$

73 The graph shown is symmetric about the y -axis. Is the function even, odd, or neither?

- a. Odd
- b. Neither
- c. Cannot be determined
- d. Even ✓

Symmetry about the y -axis is exactly the definition of an even function.

MCQ — Increasing, decreasing, and average rate of change

74 Find the average rate of change of $f(x) = x^2$ from $x = 2$ to $x = 5$.

- a. 3
- b. 7 ✓
- c. 21
- d. 29

$$\frac{f(5) - f(2)}{5 - 2} = \frac{25 - 4}{3} = 7.$$

75 Find the average rate of change of $f(x) = 3x^2 - 2x$ from $x = 0$ to $x = 4$.

- a. 40
- b. 8.5
- c. 10 ✓
- d. 46

$$f(4) = 40, f(0) = 0; \text{ rate} = \frac{40}{4} = 10.$$

76 State the interval on which $f(x) = (x - 3)^2$ is decreasing.

a. $(-\infty, 3)$ ✓

b. $(-\infty, -3)$

c. $(-3, \infty)$

d. $(3, \infty)$

The parabola's vertex is at $x = 3$; it decreases for $x < 3$.

77 Find the average rate of change of $f(x) = x^3$ from $x = 1$ to $x = 3$.

a. 8

b. 26

c. 9

d. 13 ✓

$$f(3) = 27, f(1) = 1; \text{rate} = \frac{26}{2} = 13.$$

MCQ — Operations on functions

78 Let $f(x) = x + 5$ and $g(x) = x^2 - 2$. Find $(f + g)(x)$.

a. $x^2 - x + 3$

b. $x^2 + x + 3$ ✓

c. $x^2 + x + 7$

d. $x^2 + 2x + 3$

$$(x + 5) + (x^2 - 2) = x^2 + x + 3.$$

79 Using the same f and g , find $(f \cdot g)(3)$.

a. 56 ✓

b. 48

c. 62

d. 50

$$f(3) = 8, g(3) = 7: 8 \times 7 = 56.$$

80 Using the same f and g , find $(f/g)(4)$.

a. $\frac{7}{14}$

b. $\frac{9}{16}$

c. $\frac{9}{14}$ ✓

d. $\frac{14}{9}$

$$f(4) = 9, g(4) = 14: \frac{9}{14}.$$

81 Let $f(x) = 2x$ and $g(x) = x + 6$. Find $(f - g)(x)$.

a. $3x + 6$

b. $x - 6$ ✓

c. $x + 6$

d. $-x - 6$

$$2x - (x + 6) = x - 6.$$

MCQ — Reading a function graph

82 The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.

a. -9

b. 6

c. 9

d. 0 ✓

The curve crosses the x -axis at $x = 3$, so $f(3) = 0$.

83 Using the same graph, find the values of x where $f(x) = 0$ (the x -intercepts).

a. $x = 0$ and $x = 9$

b. $x = -9$ and $x = 9$

c. $x = 3$ only

d. $x = -3$ and $x = 3$ ✓

The curve crosses the x -axis at $x = -3$ and $x = 3$.

84 Using the same graph, find the y -intercept.

a. -3

b. 9

c. 0

d. -9 ✓

At $x = 0$, the curve crosses the y -axis at $y = -9$.

MCQ — Word problems: functions in context

- 85 A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 dollars ✓

b. 150 dollars

c. 600 dollars

d. 900 dollars

$$P(15) = -2(225) + 1,200 - 300 = 450 \text{ dollars.}$$

- 86 The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.

a. 72°F

b. 68°F ✓

c. 36°F

d. 52°F

$$F(20) = \frac{9(20)}{5} + 32 = 36 + 32 = 68.$$

- 87 A ball's height (in meters) t seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.

a. 2

b. 18

c. 22 ✓

d. 42

$$h(2) = -5(4) + 40 + 2 = 22 \text{ meters.}$$

Multiple choice — easy

88 If $f(x) = x^2 - 6x - 7$, find $f(6)$.

a. -7 ✓

b. 7

c. -10

d. -4

$$f(6) = (6)^2 - 36 - 7 = -7.$$

89 If $f(x) = -3x^2 - 5x + 9$, find $f(-3)$.

a. 0

b. -6

c. -3 ✓

d. 3

$$f(-3) = -3(-3)^2 + 15 + 9 = -3.$$

90 If $f(x) = -x^2 + 7x + 1$, find $f(-1)$.

a. -4

b. 7

c. -7 ✓

d. -10

$$f(-1) = -(-1)^2 - 7 + 1 = -7.$$

91 If $f(x) = -3x^2 - x - 1$, find $f(4)$.

a. -56

b. 53

c. -50

d. -53 ✓

$$f(4) = -3(4)^2 - 4 - 1 = -53.$$

92 If $f(x) = x^2 - 5x - 6$, find $f(-3)$.

a. 21

b. 15

c. -18

d. 18 ✓

$$f(-3) = (-3)^2 + 15 - 6 = 18.$$

93 If $f(x) = x^2 + 8x - 1$, find $f(-5)$.

a. -19

b. -16 ✓

c. -13

d. 16

$$f(-5) = (-5)^2 - 40 - 1 = -16.$$

94 If $f(x) = 2x^2 - 2x + 3$, find $f(-4)$.

a. 40

b. 43 ✓

c. 46

d. -43

$$f(-4) = 2(-4)^2 + 8 + 3 = 43.$$

95 If $f(x) = 2x^2 + 8x + 5$, find $f(-1)$.

- a. -4
- b. -1 ✓
- c. 1
- d. 2

$$f(-1) = 2(-1)^2 - 8 + 5 = -1.$$

96 If $f(x) = -3x^2 - 8x + 3$, find $f(-1)$.

- a. 8 ✓
- b. 11
- c. -8
- d. 5

$$f(-1) = -3(-1)^2 + 8 + 3 = 8.$$

97 If $f(x) = 2x^2 + x + 8$, find $f(3)$.

- a. -29
- b. 32
- c. 26
- d. 29 ✓

$$f(3) = 2(3)^2 + 3 + 8 = 29.$$

98 If $f(x) = x^2 + 6x + 9$, find $f(-5)$.

- a. 4 ✓
- b. 1
- c. 7
- d. -4

$$f(-5) = (-5)^2 - 30 + 9 = 4.$$

99 If $f(x) = -3x^2 + 6x + 1$, find $f(-1)$.

a. -8 ✓

b. 8

c. -11

d. -5

$$f(-1) = -3(-1)^2 - 6 + 1 = -8.$$

100 If $f(x) = 3x^2 - 4$, find $f(-4)$.

a. 41

b. 47

c. -44

d. 44 ✓

$$f(-4) = 3(-4)^2 - 4 = 44.$$

101 If $f(x) = x^2 - 8x - 3$, find $f(5)$.

a. -21

b. 18

c. -18 ✓

d. -15

$$f(5) = (5)^2 - 40 - 3 = -18.$$

102 If $f(x) = x^2 - 2x - 2$, find $f(-2)$.

- a. 9
- b. -6
- c. 3
- d. 6 ✓

$$f(-2) = (-2)^2 + 4 - 2 = 6.$$

Multiple choice — medium

103 State the domain of $f(x) = \sqrt{-x - 6}$.

- a. $x \leq -6$ ✓
- b. $x \leq -4$
- c. $x \leq -8$
- d. $x \geq -6$

$$-x - 6 \leq 0 \Rightarrow x \leq -6.$$

104 State the domain of $f(x) = \frac{1}{x - 9}$.

- a. $x \neq 7$
- b. $x = 9$
- c. $x \neq 9$ ✓
- d. $x \neq 11$

All real numbers except $x = 9$ (where the denominator is zero).

105 State the domain of $f(x) = \frac{1}{x-7}$.

- a. $x \neq 5$
- b. $x = 7$
- c. $x \neq 9$
- d. $x \neq 7$ ✓

All real numbers except $x = 7$ (where the denominator is zero).

106 State the domain of $f(x) = \frac{1}{x-11}$.

- a. $x \neq 9$
- b. $x \neq 13$
- c. $x = 11$
- d. $x \neq 11$ ✓

All real numbers except $x = 11$ (where the denominator is zero).

107 State the domain of $f(x) = \sqrt{2x-2}$.

- a. $x \geq -1$
- b. $x \geq 3$
- c. $x \leq 1$
- d. $x \geq 1$ ✓

$$2x - 2 \geq 0 \Rightarrow x \geq 1.$$

108 State the domain of $f(x) = \sqrt{3x+6}$.

- a. $x \geq -2$ ✓
- b. $x \geq -4$
- c. $x \leq -2$
- d. $x \geq 0$

$$3x + 6 \geq 0 \Rightarrow x \geq -2.$$

109 State the domain of $f(x) = \sqrt{-3x + 24}$.

a. $x \leq 8$ ✓

b. $x \leq 6$

c. $x \leq 10$

d. $x \geq 8$

$$-3x + 24 \leq 0 \Rightarrow x \leq 8.$$

110 State the domain of $f(x) = \sqrt{-2x + 4}$.

a. $x \leq 4$

b. $x \geq 2$

c. $x \leq 2$ ✓

d. $x \leq 0$

$$-2x + 4 \leq 0 \Rightarrow x \leq 2.$$

111 State the domain of $f(x) = \frac{1}{x + 3}$.

a. $x \neq -1$

b. $x \neq -5$

c. $x \neq -3$ ✓

d. $x = -3$

All real numbers except $x = -3$ (where the denominator is zero).

112 State the domain of $f(x) = \sqrt{x - 2}$.

a. $x \leq 2$

b. $x \geq 4$

c. $x \geq 0$

d. $x \geq 2$ ✓

$$x - 2 \geq 0 \Rightarrow x \geq 2.$$

113 State the domain of $f(x) = \frac{1}{x + 5}$.

a. $x \neq -5$ ✓

b. $x \neq -7$

c. $x = -5$

d. $x \neq -3$

All real numbers except $x = -5$ (where the denominator is zero).

114 State the domain of $f(x) = \frac{1}{x - 1}$.

a. $x = 1$

b. $x \neq 1$ ✓

c. $x \neq -1$

d. $x \neq 3$

All real numbers except $x = 1$ (where the denominator is zero).

115 State the domain of $f(x) = \sqrt{-3x - 9}$.

a. $x \leq -1$

b. $x \leq -3$ ✓

c. $x \geq -3$

d. $x \leq -5$

$$-3x - 9 \leq 0 \Rightarrow x \leq -3.$$

116 State the domain of $f(x) = \sqrt{2x - 8}$.

a. $x \geq 2$

b. $x \geq 4$ ✓

c. $x \leq 4$

d. $x \geq 6$

$$2x - 8 \geq 0 \Rightarrow x \geq 4.$$

117 State the domain of $f(x) = \sqrt{-2x + 14}$.

a. $x \leq 7$ ✓

b. $x \leq 5$

c. $x \geq 7$

d. $x \leq 9$

$$-2x + 14 \leq 0 \Rightarrow x \leq 7.$$

Multiple choice — hard

118 Let $f(x) = -x + 4$ and $g(x) = x^2 - 5$. Find $(f \circ g)(0)$.

a. 6

b. 9 ✓

c. -9

d. 12

$$g(0) = (0)^2 - 5 = -5. \quad f(-5) = -(-5) + 4 = 9.$$

119 Let $f(x) = 3x + 2$ and $g(x) = 2x^2 + 3$. Find $(f \circ g)(0)$.

a. 8

b. -11

c. 11 ✓

d. 14

$$g(0) = 2(0)^2 + 3 = 3. f(3) = 3(3) + 2 = 11.$$

120 Let $f(x) = -2x + 8$ and $g(x) = -x^2$. Find $(f \circ g)(4)$.

a. 40 ✓

b. -40

c. 43

d. 37

$$g(4) = -(4)^2 = -16. f(-16) = -2(-16) + 8 = 40.$$

121 Let $f(x) = -x - 2$ and $g(x) = -x^2 + 3$. Find $(f \circ g)(1)$.

a. -4 ✓

b. -7

c. -1

d. 4

$$g(1) = -(1)^2 + 3 = 2. f(2) = -(2) - 2 = -4.$$

122 Let $f(x) = x - 6$ and $g(x) = 2x^2 - 3$. Find $(f \circ g)(2)$.

a. -4

b. -1 ✓

c. 1

d. 2

$$g(2) = 2(2)^2 - 3 = 5. f(5) = (5) - 6 = -1.$$

123 Find the inverse of $f(x) = 5x - 8$.

a. $\frac{x-8}{5}$

b. $\frac{x+8}{-5}$

c. $\frac{x+8}{5}$ ✓

d. $5x - 8$

$$y = 5x - 8 \Rightarrow x = 5y - 8 \Rightarrow y = \frac{x+8}{5}. \text{ So } f^{-1}(x) = \frac{x+8}{5}.$$

124 Find the inverse of $f(x) = 3x + 5$.

a. $3x + 5$

b. $\frac{x-5}{3}$ ✓

c. $\frac{x+5}{3}$

d. $\frac{x-5}{-3}$

$$y = 3x + 5 \Rightarrow x = 3y + 5 \Rightarrow y = \frac{x-5}{3}. \text{ So } f^{-1}(x) = \frac{x-5}{3}.$$

125 Find the inverse of $f(x) = 4x - 5$.

a. $\frac{x-5}{4}$

b. $\frac{x+5}{-4}$

c. $\frac{x+5}{4}$ ✓

d. $4x - 5$

$$y = 4x - 5 \Rightarrow x = 4y - 5 \Rightarrow y = \frac{x+5}{4}. \text{ So } f^{-1}(x) = \frac{x+5}{4}.$$

126 Find the inverse of $f(x) = 4x - 3$.

a. $\frac{x-3}{4}$

b. $4x - 3$

c. $\frac{x+3}{-4}$

d. $\frac{x+3}{4}$ ✓

$$y = 4x - 3 \Rightarrow x = 4y - 3 \Rightarrow y = \frac{x+3}{4}. \text{ So } f^{-1}(x) = \frac{x+3}{4}.$$

127 Find the inverse of $f(x) = 5x + 1$.

a. $\frac{x-1}{5}$ ✓

b. $\frac{x-1}{-5}$

c. $\frac{x+1}{5}$

d. $5x + 1$

$$y = 5x + 1 \Rightarrow x = 5y + 1 \Rightarrow y = \frac{x-1}{5}. \text{ So } f^{-1}(x) = \frac{x-1}{5}.$$

128 The graph of $y = x^2$ is shifted 4 units right and 2 units down. Write the new equation.

a. $y = (x+4)^2 - 2$

b. $y = (x+4)^2 + 2$

c. $y = (x-4)^2 - 2$ ✓

d. $y = (x-4)^2 + 2$

$$y = (x-4)^2 - 2.$$

129 The graph of $y = |x|$ is shifted 3 units right and 2 units down. Write the new equation.

a. $y = |x+3| + 2$

b. $y = |x+3| - 2$

c. $y = |x-3| - 2$ ✓

d. $y = |x-3| + 2$

$$y = |x-3| - 2.$$

130 The graph of $y = x^2$ is shifted 1 unit left and 4 units down. Write the new equation.

a. $y = (x - 1)^2 + 4$

b. $y = (x + 1)^2 + 4$

c. $y = (x + 1)^2 - 4$ ✓

d. $y = (x - 1)^2 - 4$

$y = (x + 1)^2 - 4.$

131 The graph of $y = |x|$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = |x + 7| + 3$

b. $y = |x - 7| - 3$

c. $y = |x + 7| - 3$ ✓

d. $y = |x - 7| + 3$

$y = |x + 7| - 3.$

132 The graph of $y = x^2$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = (x - 7)^2 + 3$

b. $y = (x - 7)^2 - 3$

c. $y = (x + 7)^2 + 3$

d. $y = (x + 7)^2 - 3$ ✓

$y = (x + 7)^2 - 3.$